

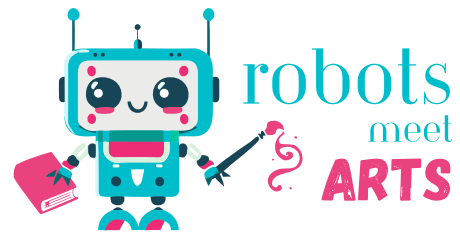


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HUMAN ROBOT PREPARES BREAKFAST.



**Subject:**

introduction

Content:

A human is also programmed.

**Computational practices:
choose what fits**

- **Decomposing**
- Pattern recognition
- Abstraction
- **Algorithms**
- **Problem solving**

Unplugged/plugged:

**Preparation time:**

10 min

**Age:**

8-12 years

**Duration:**

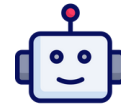
25 min

**Location:**

classroom

**Group distribution:**

classroom and small groups

**Difficulty 1/3****After this lesson, the children will be able to...**

- understanding what unplugged programming entails.
- thinking logically and practising algorithms without using computer.
- understand the importance of sequencing actions to achieve a given goal.
- understand the importance of clear and unambiguous instructions.

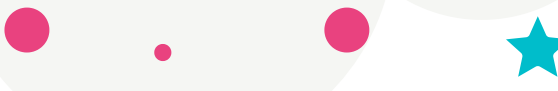


Materials needed:

- worksheet per group of pupils
- 1 bowl
- 1 spoon
- cereal
- milk

Helpful links:

https://petranmeertens.nl/?page_id=1172 (video in Dutch)



Preparation and teacher notes

For each 3 pupils, copy the attached worksheet with instructions.

They front the bowl, spoon, cereal and milk ready.

If you have a strong group, you can also choose to omit the instruction sheet and let the pupils write down instructions for the human robot on a blank paper themselves using their own words.

You can also choose not to work with real food, but to use (Lego) blocks and water instead.

Warm up 5'

When getting ready for school, who still thinks about the different actions you have to do? In what order should you get dressed? First your jumper and then your T-shirt? Or rather the other way around?

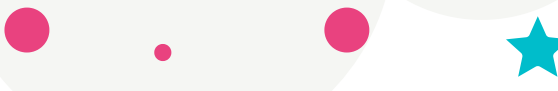
Today we are going to think about the unconscious steps we take to prepare breakfast anyway. We are going to eat a bowl of cereal with milk.

What steps should you take to do this?

First think about it in a group of 3 students. You may use a sheet with instructions to put in the right order.

Note! The robot will do everything literally. So it is best to try beforehand whether the instructions are correct.





Preparation and teacher notes

The teacher plays the human robot. It is fun to sound a bit like a robot. When mistakes are made, perform verbatim so students see that the sequence or chosen action has a wrong result. You can also say error when something gets stuck.

As much as possible, let the pupils think, reflect and improve themselves. Change groups regularly to give the next instruction to the human robot.

Build up 15'

In group of three, the students list the instructions for preparing breakfast.

When the students are ready, we will classically have the human robot prepare and eat cereal.

The students themselves notice when a certain step is skipped or unclear to the human robot.

Rehearsal 5'

End this activity with a short discussion. Have students reflect on their own actions. Following questions can help:

Did you expect that it would take so many steps to prepare and eat e braekfast?

What dit you find difficult / easy?

Clean up 5'





right hand



left hand

take

open

pour

put down

stir

put down

close

shake

hold on

bring

eat

turn

bowl

spoon

milk

cornflakes

cap

mouth

table

bottle

slow

fast

hard

soft

little

a lot

back

above





Have the human robot prepare a bowl of cereal for breakfast. Write down the different steps the robot has to take. You can choose words from the attached list.

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